

Anxiety-Adaptive Exoskeleton Control

Hypothesis

Anxiety-adaptive control + immunization training → stronger and more durable embodiment and trust than fixed-assistance or uncontrolled failure exposure.

Anxiety Model

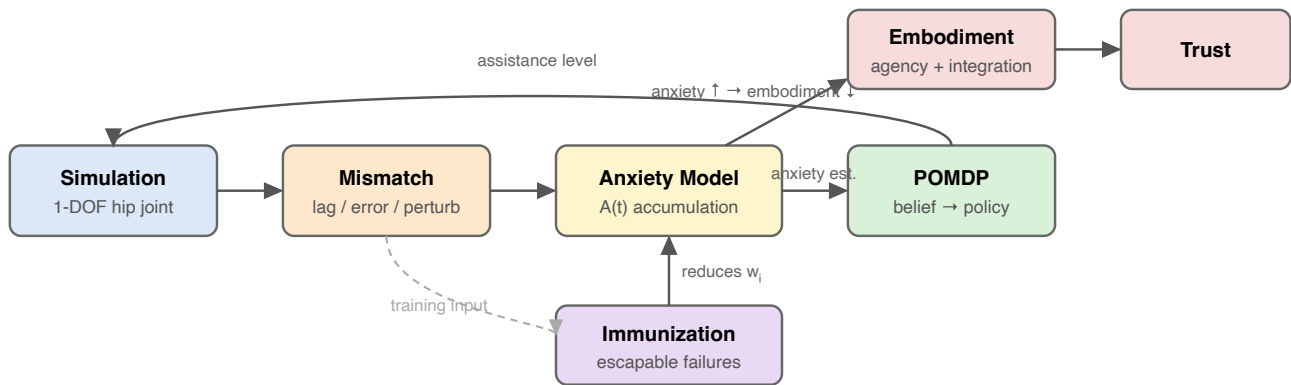
Proposed model — based on stress accumulation theory, HRV recovery literature, and predictive processing (not empirically fitted):

$$A(t) = A_{\text{baseline}} + \sum_i w_i \cdot S_i \cdot e^{-\lambda(t-t_i)}$$

- Stressor weights: $w_{\text{temporal}} = 0.5$, $w_{\text{magnitude}} = 1.0$, $w_{\text{perturbation}} = 2.0$ (*Fine tuning is needed*)
 - Decay $\lambda = 0.03 \text{ s}^{-1} \rightarrow$ half-life $\approx 23\text{s}$ (matches HRV recovery timescale, *Fine tuning is needed*)
 - Immunization reduces effective stressor weight through repeated controllable exposure
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Approach: Computational Simulation

- 1-DOF hip joint pendulum (human torque + PD exoskeleton torque)
- Mismatch injected deliberately: temporal lag, magnitude error, perturbation
- POMDP observes noisy proxies (HRV, tracking error, smoothness) → adjusts assistance level



Experiments

1. Fixed vs. Adaptive Assistance

- Fixed: constant 75% assistance under mismatch
- Adaptive: POMDP reduces assistance when anxiety high → restores agency
- Expected: lower mean anxiety, minor tracking error increase

2. Immunization Protocol

- Trained group: brief *escapable* failures before the main test
- Both groups face the same large uncontrollable perturbation
- Expected: lower peak anxiety, faster recovery in trained group

3. Sensitivity Analysis

- Vary decay rate, assistance thresholds, mismatch type
- Confirm robustness to parameter changes

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